

MDGen: Generative Molecular Dynamics Simulation

A Comprehensive Technical Survey
with Background, Prior Work, Architecture and Analysis

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Abstract

MDGen is the first generative model to produce temporally coherent, full-backbone protein trajectories at $\sim 10,000\times$ the speed of explicit-solvent molecular dynamics (MD). This survey provides a comprehensive technical exposition covering protein biophysics, denoising diffusion probabilistic models on non-Euclidean manifolds ($SO(3)$, T^4), $SE(3)$ -equivariance, and a complete landscape of prior work—Normal Mode Analysis, Boltzmann generators, and independent-frame diffusion models. The MDGen architecture alternates $SE(3)$ -equivariant Invariant Point Attention (IPA) with temporal axial attention, enabling joint modelling of equilibrium thermodynamics and kinetic temporal structure. Trained on ATLAS (1,400 proteins $\times$ 300 ns each), MDGen achieves RMSF Pearson r approx 0.87, Ramachandran JSD approx 0.08, and correctly reproduces autocorrelation functions and frame-to-frame step-size distributions.

1 Introduction

Proteins are the primary molecular machines of living systems. They catalyse chemical reactions, transmit cellular signals, provide structural integrity, and execute the vast majority of biochemical functions essential for life. A protein's biological activity depends critically not only on its average three-dimensional structure but on its dynamical behaviour: the ensemble of conformations it explores at physiological temperature, the timescales of interconversion between states, and the kinetic pathways connecting them (Adcock & McCammon, 2006).

Despite decades of progress in experimental structure determination—X-ray crystallography, cryo-EM, NMR spectroscopy—these methods report static or ensemble-averaged snapshots. Molecular dynamics (MD) simulation fills this gap by numerically integrating Newton's equations of motion for every atom, generating detailed time-series of atomic positions. Modern GPU-accelerated MD packages can simulate a 100-residue protein at $\sim 100\text{-}200$ ns per day. Yet biologically important timescales—domain motions, loop gating, folding transitions—occur on microsecond to millisecond timescales, spanning 9-12 orders of magnitude above the obligatory 2 fs integration time step. This timescale gap makes MD prohibitively expensive for large-scale ensemble characterisation or drug discovery campaigns.

Core motivation. MDGen asks: can a neural network, trained on a large corpus of MD trajectories, learn to generate new trajectory segments that faithfully reproduce both the equilibrium thermodynamics AND the kinetic temporal structure of real MD—without running the simulation itself?

All prior generative models for protein conformations operate in an i.i.d. sampling regime: each sample is drawn independently from an approximation of the equilibrium distribution. Such models cannot reproduce temporal autocorrelation functions, kinetically realistic step sizes between consecutive frames, or any observable that depends on the ordering of configurations in time. MDGen (Jing et al., NeurIPS 2024) addresses this gap by posing the generative problem at the level of entire trajectory segments: given a protein sequence, generate T consecutive backbone frames whose joint

distribution matches that of a real MD simulation. The key technical contribution is an architecture alternating SE(3)-equivariant Invariant Point Attention (IPA) with novel temporal axial attention. Trained on ATLAS (1,400 proteins x 300 ns each), MDGen achieves RMSF Pearson $r \sim 0.87$ and correct temporal kinetics at $\sim 10,000\times$ the speed of MD.

2 Biological and Physical Background

2.1 Proteins: Structure and Function

A protein is a linear polymer of amino acid residues connected by peptide bonds. Twenty standard amino acids, differing only in their side chain, are encoded by the genetic code. Proteins range from tens to thousands of residues; a typical globular protein has 100-500 residues.

The structural hierarchy is decomposed into four levels:

- **Primary structure:**

Primary structure: the amino-acid sequence, a string over a 20-letter alphabet.

- **Secondary structure:**

Secondary structure: local backbone regularities—alpha-helices (right-handed coils stabilised by i to $i+4$ hydrogen bonds) and beta-strands (extended chains forming hydrogen-bonded beta-sheets).

- **Tertiary structure:**

Tertiary structure: the complete 3D fold of a single polypeptide chain, governed by hydrophobic packing, electrostatics, hydrogen bonding, and van der Waals interactions.

- **Quaternary structure:**

Quaternary structure: assembly of multiple polypeptide chains into a complex.

The protein backbone consists of the repeating N-Ca-C=O unit running through every residue. Backbone conformation is almost completely determined by two dihedral angles: phi (N-Ca bond rotation) and psi (Ca-C bond rotation). MDGen operates exclusively on backbone heavy atoms, representing each residue as a rigid frame in SE(3), with side chains encoded separately as torsion angles χ_1 to χ_4 .

2.2 The Protein Energy Landscape and Conformational Ensembles

A protein in solution at temperature T does not adopt a single static conformation. Instead, it continuously fluctuates according to Boltzmann statistics. The probability of finding the system in conformation x is:

$$p(x) \propto \exp(-E(x)/k_B T) \quad (1)$$

where $E(x)$ is the potential energy, k_B is Boltzmann's constant, and $k_B T \sim 0.6$ kcal/mol at 300 K (Frauenfelder et al., 1991). Key features of the energy landscape relevant to MDGen:

- Multiple minima: most functional proteins populate more than one conformation (e.g., open/closed enzyme active sites).
- Hierarchical roughness: fast local motions (10 fs) nested within slower large-scale rearrangements (us-ms).
- Anharmonicity: large-amplitude motions cannot be described by a harmonic approximation—a fundamental limitation of Normal Mode Analysis.

2.3 Molecular Dynamics Simulation

MD simulation numerically integrates Newton's equations of motion:

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_{\mathbf{r}_i} E(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad (2)$$

The empirical force field E comprises bonded terms (harmonic bond stretches, angle bends, periodic dihedral torsions) and non-bonded terms (Lennard-Jones 6-12 van der Waals, Coulomb electrostatics via Particle Mesh Ewald). Widely used force fields include AMBER ff14SB (used for ATLAS), CHARMM36m, and OPLS-AA.

Timescale problem. The fastest atomic vibrations (~ 10 fs) impose an integration ceiling of $dt = 2$ fs. Observing biologically important motions requires 10^8 to 10^{11} steps (100 ns to 100 μ s). A single microsecond simulation of a 100-residue protein on an A100 GPU cluster takes ~ 1 -2 weeks. Screening thousands of variants, as required in drug discovery, is practically infeasible with standard MD.

2.4 Trajectory Representation for MDGen

An MD trajectory is a time-ordered sequence of frames ($\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$), where each frame $\mathbf{x}_t$ specifies the full atomic coordinates at time t . In MDGen, the relevant degrees of freedom per residue i at time t are a rigid body frame in $SE(3)$ plus torsion angles on the 4-torus:

$$\mathbf{x} = \{(R_i, \mathbf{t}_i, \chi_i)\}_{i=1}^N \in SE(3)^N \times \mathbb{T}^{4N} \quad (3)$$

A trajectory window of T frames is the joint object $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$. Training uses $T = 16$ frames, corresponding to ~ 16 ns at the ATLAS 1 ns frame rate.

3 Theoretical Background: Diffusion Models

3.1 Denoising Diffusion Probabilistic Models (DDPMs)

DDPMs (Ho et al., 2020) define a generative model through two coupled Markov chains: a forward process that gradually destroys structure by adding Gaussian noise, and a reverse process learned to reconstruct data from noise.

Forward process. Given clean data $\mathbf{x}_0$, the forward process is:

$$q(\mathbf{x}_s | \mathbf{x}_{s-1}) = \mathcal{N}(\mathbf{x}_s; \sqrt{1 - \beta_s} \mathbf{x}_{s-1}, \beta_s \mathbf{I}) \quad (4)$$

The marginal at step s has a closed form with $\alpha_s = \prod_{k=1}^s (1 - \beta_k)$:

$$q(\mathbf{x}_s|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_s; \sqrt{\tilde{\alpha}_s} \mathbf{x}_0, (1 - \tilde{\alpha}_s)\mathbf{I}) \quad (5)$$

As $s \rightarrow S$, the distribution approaches $\mathcal{N}(0, \mathbf{I})$ — pure isotropic noise. The training objective minimises noise prediction error:

$$\mathcal{L} = \mathbb{E}_{s, \mathbf{x}_0, \varepsilon} \|\varepsilon - \varepsilon_\theta(\sqrt{\tilde{\alpha}_s} \mathbf{x}_0 + \sqrt{1 - \tilde{\alpha}_s} \varepsilon, s)\|^2 \quad (6)$$

3.2 Score-Based Models and the Continuous-Time SDE Perspective

Song et al. (2021) cast diffusion models as continuous-time stochastic differential equations (SDEs). The forward SDE is:

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{W} \quad (7)$$

where $\mathbf{W}$ is a standard Brownian motion. The reverse SDE (Anderson, 1982) runs backwards from noise to data:

$$d\mathbf{x} = [f - g^2 \nabla_{\mathbf{x}} \log p_t] dt + g d\tilde{\mathbf{W}} \quad (8)$$

The key quantity is the score function: the gradient of the log density, which points toward higher probability regions. A score network $s_\theta(\mathbf{x}, t)$ is trained via denoising score matching (Vincent, 2011). DDIM (Song et al., 2021) provides a deterministic ODE equivalent enabling high-quality sampling in ~ 50 steps instead of 500:

$$\mathbf{x}_{s-1} = \sqrt{\tilde{\alpha}_{s-1}} \left(\frac{\mathbf{x}_s - \sqrt{1 - \tilde{\alpha}_s} \varepsilon_\theta}{\sqrt{\tilde{\alpha}_s}} \right) + \sqrt{1 - \tilde{\alpha}_{s-1}} \varepsilon_\theta \quad (9)$$

3.3 Diffusion on Non-Euclidean Manifolds

Standard Gaussian diffusion assumes flat Euclidean space $\mathbb{R}^d$. Protein backbone frames involve two non-Euclidean components: rotations R_i in $\text{SO}(3)$ (a compact curved Lie group) and torsion angles in the circle S^1 (with side-chain state on the 4-torus T^4).

Diffusion on $\text{SO}(3)$. MDGen uses the Isotropic Gaussian on $\text{SO}(3)$ (IGSO3; De Bortoli et al., 2022):

$$q(R_s|R_0) = \text{IGSO3}(R_s; R_0, \sigma_s^2) \quad (10)$$

Its score uses the Riemannian gradient—the analogue of the Euclidean score on the curved $\text{SO}(3)$ manifold, computed via the heat kernel on $\text{SO}(3)$. For torsion angles θ , the correct distribution is a wrapped Gaussian:

$$q(\theta_s|\theta_0) = \sum_{k=-\infty}^{\infty} \mathcal{N}(\theta_s; \theta_0 + 2\pi k, \sigma_s^2) \quad (11)$$

3.4 SE(3) Equivariance

The group $\text{SE}(3) = \text{SO}(3) \times \mathbb{R}^3$ consists of all rotations and translations of $\mathbb{R}^3$. A map f is $\text{SE}(3)$ -equivariant if rotating/translating the input produces a correspondingly rotated/translated output:

$$f(g \cdot \mathbf{x}) = \rho_{\text{out}}(g) f(\mathbf{x}), \quad g \in \text{SE}(3) \quad (12)$$

$\text{SE}(3)$ -equivariant networks (Fuchs et al., 2020; Satorras et al., 2021) architecturally enforce this constraint exactly. The physics of a protein does not change if we rotate or translate the entire coordinate system; a model violating this symmetry must learn it from data, wasting capacity.

3.5 Backbone Frame Representation

AlphaFold2 (Jumper et al., 2021) introduced the backbone frame representation that MDGen inherits. Each residue i is assigned a rigid body in $\text{SE}(3)$: a translation $\mathbf{t}_i$ (C_α position) and a rotation $\mathbf{R}_i$ (local backbone orientation defined by N , C_α , C atoms). Together, $\mathbf{T}_i = (\mathbf{R}_i, \mathbf{t}_i)$ defines a frame in which the local chemical environment appears in a standardised orientation.

4 Prior Work and Its Limitations

Understanding MDGen's novelty requires surveying the space of prior methods for computational characterisation of protein conformational dynamics, organised into four categories.

4.1 Normal Mode Analysis (NMA)

NMA (Tirion, 1996; Brooks & Karplus, 1983) approximates the protein energy landscape as a multidimensional harmonic bowl centred on a minimised structure. The equations of motion separate into $3N-6$ independent harmonic oscillators (normal modes). The lowest-frequency modes correspond to large-scale collective motions and predict B-factors and RMSF profiles.

Strengths of NMA:

- Computationally cheap: a single eigendecomposition of the Hessian.
- Provides physically interpretable collective modes.
- Reasonable agreement with MD-derived RMSF for ordered secondary structure elements.

Fundamental limitations:

- Harmonic approximation: strictly valid only near the energy minimum. Loop motions, domain swings, and barrier-crossing transitions are entirely outside the scope of the harmonic model.
- No kinetics: NMA produces i.i.d. Gaussian samples with no notion of temporal ordering or transition rates.
- Single minimum: cannot represent multi-modal distributions or conformational heterogeneity between distinct energy basins.
- RMSF Pearson $r \sim 0.64$ on ATLAS test set—substantially below MDGen's $r \sim 0.87$.

4.2 Boltzmann Generators and Flow-Based Samplers

Boltzmann generators (Noe et al., 2019) use normalising flows to learn a direct mapping between a Gaussian base distribution and the Boltzmann distribution $p(x)$ proportional to $\exp(-E(x)/k_B T)$. Samples can be drawn without simulating the dynamics.

Strengths:

- Principled probabilistic model with exact likelihood.
- Can provide reweighting factors to correct force-field imperfections.

Fundamental limitations:

- Scalability: normalising flows for all-atom configurations of 100-500 residues remain computationally intractable; most demonstrations are restricted to ≤ 50 -residue systems.
- No temporal coherence: like NMA, produces i.i.d. equilibrium samples with no mechanism for temporally ordered trajectories.
- Requires the energy function: training evaluates $E(x)$, coupling the method to the expensive force field evaluation.

4.3 Independent-Frame Protein Diffusion Models

A family of recent diffusion models generates protein structures by reversing a noising process on backbone frames, producing independent equilibrium samples.

- **FrameDiff:**

FrameDiff (Yim et al., ICML 2023): diffuses over $SE(3)^N$ backbone frames; samples drawn independently.

- **EigenFold:**

EigenFold (Jing et al., 2023): decomposes protein structure along spring-network eigenmodes; coarse-to-fine generation.

- **AlphaFlow:**

AlphaFlow (Jing et al., ICML 2024): fine-tunes AlphaFold2 as a flow-matching model conditioned on MSAs.

- **BioEmu:**

BioEmu (Lewis et al., Science 2025): large-scale diffusion model trained on MD trajectories and experimental data.

Shared limitation: ALL of the above generate INDEPENDENT conformational samples. They cannot reproduce: (1) the temporal autocorrelation function $C(\tau)$ — which drops to zero immediately at lag ≥ 1 for i.i.d. samplers; (2) physically realistic frame-to-frame step sizes (~ 0.1 - 0.3 Å RMSD); (3) transition pathway information. These are not aesthetic deficiencies—kinetic rate constants, mean first-passage times, and committed fractions are fundamentally temporal and inaccessible to equilibrium samplers.

4.4 Static Structure Prediction Models

- **AlphaFold2:**
AlphaFold2 (Jumper et al., 2021): near-experimental accuracy for the single lowest-energy structure. Produces one structure per input; not designed for ensemble generation or dynamics.
- **ESMFold:**
ESMFold (Lin et al., 2023): fast single-structure prediction via language model without MSA. MDGen uses it only for the initial conditioning structure.
- **RFdiffusion / Chroma:**
RFdiffusion (Watson et al., 2023) and Chroma (Ingraham et al., 2023): diffusion models for protein design—generating novel backbones satisfying functional constraints. Not evaluated on MD ensemble metrics.

4.5 Summary Comparison Table

Method	Equil. Ensemble	Temporal Coh	Speed vs MD	Large Proteins
MD simulation	Yes	Yes	1x (baseline)	Yes
NMA	Partial	No	~1000x	Yes
Boltzmann generators	Yes	No	~1000x	No
EigenFold / AlphaFlow	Yes	No	~10000x	Yes
BioEmu	Yes	No	~10000x	Yes
MDGen (this work)	Yes	Yes	~10000x	Yes

5 MDGen: Method

5.1 Problem Formulation

Definition (Trajectory generation). Given a protein sequence $s = (s_1, \dots, s_N)$, generate a trajectory window $X = (x_1, \dots, x_T)$ in $(SE(3)^N \times T^4N)^T$ such that the joint distribution $p_{\theta}(X|s)$ approximates the distribution over T-frame windows drawn from equilibrium MD trajectories.

This formulation requires two simultaneous properties:

- Equilibrium marginal correctness: the marginal $p(\mathbf{x}_t | s)$ for any single time t matches the MD equilibrium distribution.
- Temporal coherence: the joint $p(\mathbf{x}_t, \mathbf{x}_{t+1} | s)$ reflects the physically correct Markov transition structure of MD dynamics.

5.2 Forward Diffusion over Trajectory Space

MDGen applies independent noise to each trajectory frame, keeping the forward process factorised over the time axis:

$$q(\mathbf{X}^{(s)} | \mathbf{X}^{(0)}) = \prod_{t=1}^T q(\mathbf{x}_t^{(s)} | \mathbf{x}_t^{(0)}) \quad (13)$$

Within each frame, diffusion is factorised over geometric components (translation, rotation, torsion angles):

$$q(\mathbf{x}_t^{(s)} | \mathbf{x}_t^{(0)}) = q_{\mathbb{R}^3}(\mathbf{t}^{(s)} | \mathbf{t}^{(0)}) \cdot q_{SO(3)}(R^{(s)} | R^{(0)}) \cdot q_T(\chi^{(s)} | \chi^{(0)}) \quad (14)$$

Key insight: frames are noised independently in the forward process, but the reverse process denoises ALL T frames jointly. Temporal correlations are entirely encoded in the LEARNED denoising network—specifically in the temporal attention layers.

5.3 Neural Architecture

5.3.1 Sequence Encoder: ESM-2

The protein sequence s is embedded by the pre-trained protein language model ESM-2 (Lin et al., 2023) into per-residue feature vectors $\mathbf{h}_i \in \mathbb{R}^d$. ESM-2 is kept frozen during MDGen training. Its representations capture evolutionary and physicochemical information, projected to internal dimension d via a learned linear layer.

5.3.2 Spatial Attention: Invariant Point Attention (IPA)

At each trajectory frame t , inter-residue information exchange is mediated by Invariant

Point Attention (IPA; Jumper et al., 2021). IPA augments standard multi-head attention with 3D query/key point pairs expressed in each residue's local backbone frame:

$$a_{ij} = \text{softmax}_j \left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} - \frac{\gamma}{2} \sum_h \|T_i \mathbf{p}_i^h - T_j \mathbf{p}_j^h\|^2 \right) \quad (15)$$

Here $T_i = (R_i, t_i)$ maps local frame points to global coordinates, $\mathbf{p}_i^h$ are learned 3D query points in residue i 's local frame, and $\gamma > 0$ is a learnable scalar. The attention is SE(3)-invariant: the point-distance term is unchanged by any global rotation or translation. Two residues distant in sequence but close in 3D space receive a high attention weight—enabling the model to reason about spatial contacts across the structure.

5.3.3 Temporal Attention: Axial Attention Across Time

The defining architectural innovation in MDGen is axial temporal attention (Ho et al., 2019). After the spatial IPA pass (across residues at fixed frame t), a temporal attention pass operates across all T time steps for each residue i :

$$\mathbf{h}_{i,t}^{\text{new}} = \sum_{s=1}^T \alpha_{ts} \mathbf{v}_{i,s}, \quad \alpha_{ts} = \text{softmax}_s \left(\frac{\mathbf{q}_{i,t} \cdot \mathbf{k}_{i,s}}{\sqrt{d}} \right) \quad (16)$$

This allows every time step to attend to every other time step within the trajectory window. The loss penalises temporally incoherent predictions, and the temporal attention weights adapt to propagate continuity information along the time axis. The spatial and temporal attention layers alternate in a stack of L blocks:

- IPA_l (spatial pass): all N residues exchange information at each fixed frame t .
- TempAttn_l (temporal pass): each residue i attends across frames $t = 1, \dots, T$.
- FFN_l: position-wise feed-forward network.

5.3.4 Score Head and Model Size

After L alternating blocks, separate prediction heads convert the latent representation into predicted noise for each geometric component: ϵ_{trans} in $\mathbb{R}^3$, ϵ_{rot} in $\text{so}(3)$, ϵ_{tors} in $\mathbb{R}^4$. The rotation score is expressed as a vector in the Lie algebra $\text{so}(3) \sim \mathbb{R}^3$. Total parameter count: approximately 100M.

5.4 Training Objective

The training loss is a weighted sum of denoising score matching losses for each geometric component, plus a structural violation penalty:

$$\mathcal{L} = \mathbb{E} [\lambda_t \mathcal{L}_{\mathbb{R}^3} + \lambda_r \mathcal{L}_{SO(3)} + \lambda_x \mathcal{L}_{\mathcal{T}} + \lambda_v \mathcal{L}_{viol}] \quad (17)$$

- $\mathcal{L}_{trans}$: standard MSE on C_{α} displacements. Weight $\lambda_t = 1.0$.
- $\mathcal{L}_{rot}$: Riemannian score matching loss on the $SO(3)$ manifold. Weight ~ 1.0 .
- $\mathcal{L}_{tors}$: wrapped Gaussian score matching for torsion angles. Weight ~ 0.5 .
- $\mathcal{L}_{viol}$: steric clash penalty and backbone bond-geometry violations. Weight ~ 0.01 .

All N residues and all $T = 16$ frames contribute to each gradient update. Diffusion noise level $s \sim \text{Uniform}(0, S)$ with $S = 500$. Optimiser: AdamW, $lr = 1e-4$, cosine annealing, batch size ~ 64 windows, on 8 x A100 GPUs.

5.5 Inference: Sampling

Sampling runs the reverse diffusion starting from pure noise for all T frames simultaneously: translations $\sim N(0, I)$, rotations $\sim \text{Uniform}(SO(3))$, torsions $\sim \text{Uniform}(T^4)$. MDGen supports two modes:

- DDPM (500 steps): stochastic reverse process; highest sample quality; $\sim$ minutes per trajectory.
- DDIM (50 steps): deterministic ODE; 10x faster than DDPM with minimal quality loss; $\sim$ seconds per trajectory.

Both modes produce trajectories approximately 10,000x faster than equivalent explicit-solvent MD simulation.

5.6 Trajectory Extension via Diffusion Inpainting

MDGen can generate trajectories conditioned on a known conformation $x_1^{(0)}$ (e.g., a crystal structure) using diffusion inpainting (Lugmayr et al., 2022). During reverse diffusion, the first-frame variable is periodically replaced with its correctly-noised

counterpart:

$$\mathbf{x}_1^{(s)} \leftarrow \sqrt{\tilde{\alpha}_s} \mathbf{x}_1^{(0)} + \sqrt{1 - \tilde{\alpha}_s} \varepsilon \quad (18)$$

Interpolation mode: conditioning on BOTH $\mathbf{x}_1^{(0)}$ and $\mathbf{x}_T^{(0)}$ generates a plausible transition pathway between two known protein conformations (e.g., open $\leftrightarrow$ closed states of adenylate kinase). This has no equivalent in any prior independent-frame sampler.

6 Training Data: The ATLAS Database

MDGen is trained on the ATLAS (Atlas of Trajectory Landscapes and Associated Simulations) database (Vander Meersche et al., Nucleic Acids Research, 2024), the largest open repository of protein MD trajectories assembled to support machine learning on protein dynamics.

Simulation protocol:

- Force field: AMBER ff14SB with explicit TIP3P water, NPT ensemble, 300 K, 1 atm.
- Three independent 100 ns replicas per protein, yielding 300 ns total.
- Frames saved every 1 ns (~ 300 frames per protein).
- Simulation time step: 2 fs with SHAKE hydrogen-bond constraints.

Dataset statistics:

- $\sim 1,400$ diverse proteins from the PDB: alpha, beta, alpha/beta folds, IDPs.
- Protein lengths: ~ 50 to ~ 500 residues.
- Aggregate simulation time: ~ 1.2 us across all proteins and replicas.
- Train split: $\sim 1,265$ proteins; Test split: ~ 135 held-out proteins.

During training, a random contiguous window of $T = 16$ consecutive frames is extracted from each replica trajectory per minibatch. Overlapping windows are permitted as data augmentation.

7 Evaluation Protocol and Results

7.1 Evaluation Metrics

7.1.1 Root Mean Square Fluctuation (RMSF)

RMSF quantifies per-residue positional flexibility:

$$\text{RMSF}_i = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\mathbf{r}_i(t) - \langle \mathbf{r}_i \rangle\|^2} \quad (19)$$

High RMSF = flexible region (loops, termini). Low RMSF = rigid region (helices, beta-sheets). The primary summary metric is Pearson correlation r between generated and reference MD RMSF profiles, averaged over test proteins.

7.1.2 Ramachandran Distribution: Jensen-Shannon Divergence

MDGen's generated (ϕ , ψ) backbone dihedral distribution is compared to reference MD using Jensen-Shannon Divergence (JSD), discretised over a 36x36 grid of (ϕ , ψ) bins. Lower JSD indicates better agreement.

7.1.3 Temporal Metrics (unique to MDGen)

Two metrics test temporal coherence—quantities impossible to evaluate from i.i.d. samplers:

- Frame-to-frame RMSD: in real MD, consecutive frames differ by only $\sim 0.1\text{-}0.3$ Å C $_{\alpha}$ RMSD. Independent samplers show arbitrarily large random jumps.
- Temporal autocorrelation function $C(\tau) = \langle \Delta \mathbf{r}(t) \cdot \Delta \mathbf{r}(t+\tau) \rangle / \langle |\Delta \mathbf{r}(t)|^2 \rangle$: measures structural persistence over lag τ . Physical trajectories decay smoothly from $C(0)=1$. i.i.d. samplers have $C(\tau) = 0$ for all $\tau \geq 1$ by definition.

7.2 Quantitative Results

Method	RMSF r	Ramach. JSBr	Brames RMSD	ACF	Speed
Reference MD	1.00	0.00	ground truth	ground truth	1x
MDGen (DDPM)	~0.87	~0.08	Correct	Correct	~10 ⁴ x
MDGen (DDIM)	~0.85	~0.09	Correct	Correct	~10 ⁵ x
Indep. frame diffusion	~0.71	~0.11	Wrong (random)	Zero lag	~10 ⁴ x
Normal Mode Analysis	~0.64	~0.19	Harmonic only	Harmonic only	~10 ³ x
Boltzmann generators	~0.68	~0.14	Wrong (i.i.d.)	Zero lag	~10 ³ x

MDGen achieves Pearson $r \sim 0.87$ on per-residue RMSF, outperforming both NMA ($r \sim 0.64$) and independent-frame diffusion ($r \sim 0.71$). Improvement over independent-frame diffusion confirms that temporal context improves equilibrium ensemble quality as well as kinetics. Only MDGen reproduces correct temporal autocorrelation functions and frame-to-frame step-size distributions.

7.3 Generalisation to Out-of-Distribution Fast-Folding Proteins

MDGen is evaluated on miniproteins from long reference simulations NOT in the ATLAS training set: Trp-cage (20 res.), BBA motif (28 res.), villin headpiece (35 res.), WW domain (35 res.), NTL9 (39 res.), Chignolin (10 res.). MDGen correctly identifies qualitative flexibility profiles: flexible C-terminal helices, rigid hydrophobic cores, elevated RMSF in loop regions. This demonstrates generalisation across protein fold space beyond the ATLAS training distribution.

8 Applications

8.1 Cryptic Pocket Discovery in Drug Design

Approximately 50% of protein drug targets undergo conformational changes critical for inhibitor binding. Cryptic pockets—absent in the static crystal structure but transiently open during dynamics (Oleinikovas et al., 2016)—are missed entirely by static docking. MDGen workflow:

- Generate 100-1000 backbone trajectory frames using MDGen (seconds to minutes on one GPU).
- Run pocket detection (FPOCKET, SiteMap) on each generated frame.

- Identify transiently open druggable pockets across the ensemble.
- Prioritise pockets for downstream docking and lead optimisation.

This replaces expensive multi-us MD with MDGen-generated ensembles 3-4 orders of magnitude faster. MDGen's correct temporal autocorrelation ensures conformational transitions consistent with natural protein dynamics.

8.2 Intrinsically Disordered Proteins (IDPs) and Disease

IDPs (Uversky, 2019) lack a stable folded structure and exist as dynamic conformational ensembles. They are medically critical, frequently driving aberrant aggregation in neurodegenerative diseases.

- **alpha-Synuclein:**
~140-residue IDP; conformational ensemble determines nucleation pathways of toxic amyloid fibrils in Parkinson's disease.
- **Tau:**
Highly flexible repeat domain; disordered regions initiate neurofibrillary tangle formation in Alzheimer's disease.
- **p53 (cancer):**
Most frequently mutated tumour suppressor; disordered N/C-terminal regions mediate transactivation through conformational dynamics.

AlphaFold2 reports low-confidence ($pLDDT < 50$) predictions for disordered regions—a diagnostic flag, not a structural answer. MDGen generates backbone conformational ensembles for these regions, providing a distribution of conformations rather than a misleading single coordinate set.

8.3 Ensemble-Aware Protein Engineering

- Mutational scanning: generate conformational ensembles for point mutants to predict effects on dynamics without re-running MD.
- Allosteric pathway mapping: condition on two allosteric states and use interpolation mode to identify signal-transmitting residues.
- Antibody CDR loop flexibility: characterise loop dynamics governing antigen recognition and binding affinity.

9 Limitations

- **Fixed timescale:**

$T = 16$ frames ~ 16 ns. Millisecond-scale processes (full folding, large domain translocations) are out of scope.

- **Training distribution bias:**

ATLAS predominantly contains single-chain, globular, soluble proteins (50-500 residues). Membrane proteins, large complexes, and heavily post-translationally modified proteins are under-represented.

- **Implicit solvent:**

MDGen learns from explicit-solvent MD but does not model solvent explicitly. Hydration shell dynamics and specific ion effects are captured only insofar as they manifest in backbone motions.

- **Force-field dependency:**

Generated trajectories emulate AMBER ff14SB statistics. Systematic force-field errors are inherited—MDGen is a data-driven emulator, not an independent physics engine.

- **Backbone-only representation:**

Side-chain coordinates must be reconstructed post-hoc using Rosetta Relax or DLPacker, limiting direct use in atom-level applications such as binding free energy calculations.

- **Rare barrier crossings:**

Conformational transitions requiring crossing high energy barriers (rare events on μ s+ timescales) are statistically improbable within a 16-frame window.

10 Future Directions

- **Full-atom trajectory generation:**

Jointly model backbone frames and all-atom side-chain coordinates as a unified diffusion model—enabling direct computation of binding free energies and pharmacophoric features without post-hoc reconstruction.

- **Longer timescales:**

Training on us-scale datasets (D.E. Shaw Research Anton) or enhanced-sampling trajectories (metadynamics, REST2) to unlock rarer conformational transitions.

- **Protein-ligand dynamics:**

Extending to receptor-ligand systems for ensemble-aware docking and entropic contributions to binding affinity ($\Delta G = \Delta H - T \Delta S$).

- **Experimental conditioning:**

Incorporating NMR chemical shifts, SAXS profiles, or cryo-EM density maps as soft conditioning signals during inference for physics-guided ensemble refinement.

- **Beyond proteins:**

The SE(3) trajectory diffusion framework can be adapted for RNA, DNA, carbohydrates, and lipid membranes with appropriate geometric representations.

11 Conclusion

MDGen represents a conceptual advance in generative molecular modelling. By framing the problem as trajectory generation rather than independent equilibrium sampling, MDGen is the first model to simultaneously achieve:

- High per-residue RMSF correlation with reference MD ($r \sim 0.87$ vs NMA $r \sim 0.64$).
- Physically correct Ramachandran backbone dihedral distributions (JSD ~ 0.08 vs MD reference).
- Correct temporal autocorrelation functions and frame-to-frame step-size distributions—metrics no prior independent sampler reproduces.
- Approximately 10,000x speed-up over equivalent explicit-solvent MD.

The architectural innovations—SE(3)-equivariant Invariant Point Attention (spatial) combined with temporal axial attention (kinetic)—provide a template for incorporating physical symmetries and temporal structure into generative models for other molecular and biological systems.

Central intellectual contribution: MD trajectories should be treated as the PRIMARY generative target, not a source of independent equilibrium samples. This reframing enables learning both the thermodynamics (what conformations are accessible) AND the dynamics (how fast they interconvert) from data—a template for how generative AI can emulate complex physical processes that govern molecular biology.

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